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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 02

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Deepak Rathod**
PRN No.	**23070122175**
Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	16th July, 2026
Title of Assignment:	Git Branching & Merging
Theory:	Git branching is a feature that allows developers to work on new features or bug fixes independently without affecting the main project. A branch creates a separate line of development where changes can be made safely. Once the work is completed and tested, the branch can be merged back into the main branch using the git merge command. This workflow helps in parallel development, better collaboration, and maintaining a stable main branch. **Commands Used** 1. git status – Checked the current repository status and confirmed that the working tree was clean and the local main branch was up to date. 2. git checkout -b feature-update – Created a new branch named feature-update and switched to it. 3. git branch – Displayed all local branches and verified that feature-update was the active branch. 4. git status – Checked the repository status after modifying the README.md file and confirmed that the changes were detected. 5. git add README.md – Added the modified README.md file to the staging area. 6. git commit -m "Update README on feature branch" – Committed the

staged changes to the feature-update branch with a descriptive commit message.

7. git status – Verified that the working tree was clean after committing the changes.
8. git checkout main – Switched back to the main branch.
9. type README.md *(Windows)* – Displayed the contents of the README.md file to verify that the changes from the feature branch were not yet present.
10. git merge feature-update – Merged the feature-update branch into the main branch using a fast-forward merge.
11. type README.md *(Windows)* – Verified that the merged changes were now present in the README.md file.
12. git push origin main – Pushed the updated main branch with the merged changes to the remote GitHub repository.

Implementation:

Step 1: Verify Current Repository Status

```
PS D:\CCodes\devops-lab\devops-lab> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS D:\CCodes\devops-lab\devops-lab>
```

Verified Repository Status — Checked the repository status and confirmed that the working tree was clean and the local main branch was up to date with the remote repository.

Step 2: Create and Switch to a New Branch

```
PS D:\CCodes\devops-lab\devops-lab> git checkout -b feature-update
Switched to a new branch 'feature-update'
PS D:\CCodes\devops-lab\devops-lab> git branch
* feature-update
  main
PS D:\CCodes\devops-lab\devops-lab>
```

Created Feature Branch — Created a new branch named feature-update and switched to it successfully using git checkout -b.

Step 3: Modify README.md

```
PS D:\CCodes\devops-lab\devops-lab> git status
On branch feature-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
PS D:\CCodes\devops-lab\devops-lab>
```

Modified README.md — Updated the README.md file on the feature-update branch and verified that Git detected the modified file.

Step 4: Stage and Commit the Changes

```
PS D:\CCodes\devops-lab\devops-lab> git add README.md
PS D:\CCodes\devops-lab\devops-lab> git commit -m "Update README on feature branch"
[feature-update a5a58d5] Update README on feature branch
 1 file changed, 0 insertions(+), 0 deletions(-)
PS D:\CCodes\devops-lab\devops-lab> git status
On branch feature-update
nothing to commit, working tree clean
PS D:\CCodes\devops-lab\devops-lab>
```

Committed Feature Changes — Staged and committed the updated README.md file on the feature-update branch with a descriptive commit message.

Step 5: Switch Back to Main Branch

```
PS D:\CCodes\devops-lab\devops-lab> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\CCodes\devops-lab\devops-lab> type README.md
# DevOps Lab
PS D:\CCodes\devops-lab\devops-lab>
```

Switched to Main Branch — Switched back to the main branch and verified that the feature branch changes were not yet available.

Step 6: Merge the Feature Branch

```
PS D:\CCodes\devops-lab\devops-lab> git merge feature-update
Updating f82af39..a5a58d5
Fast-forward
 README.md | Bin 30 -> 128 bytes
 1 file changed, 0 insertions(+), 0 deletions(-)
PS D:\CCodes\devops-lab\devops-lab> type README.md
# DevOps Lab

This update adds details about a new feature.
PS D:\CCodes\devops-lab\devops-lab>
```

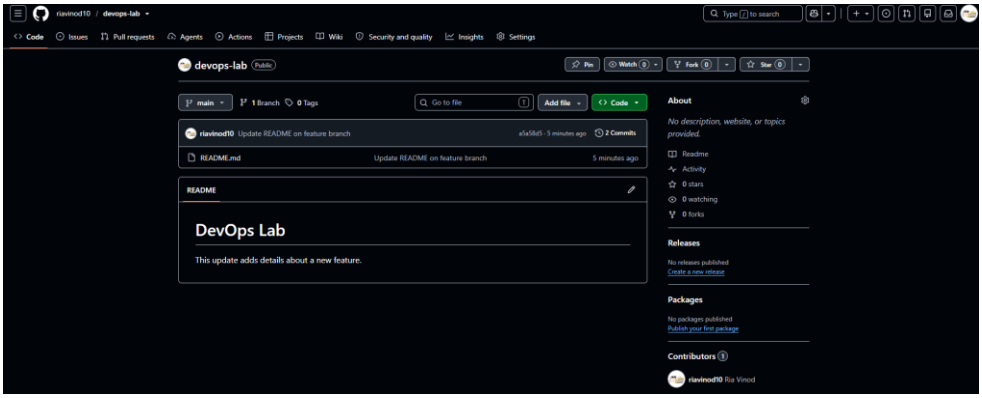
Merged Feature Branch — Successfully merged the feature-update branch into the main branch using a fast-forward merge.

Step 7: Push Updated Main Branch

```
PS D:\CCodes\devops-lab\devops-lab> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 345 bytes | 345.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/riavinod10/devops-lab
   f82af39..a5a58d5  main -> main
PS D:\CCodes\devops-lab\devops-lab>
```

Pushed Updated Main Branch — Uploaded the merged changes from the local main branch to the remote GitHub repository.

Step 8: Verify on GitHub



Repository Verified — Verified on GitHub that the merged changes are present in the main branch and that the commit history reflects both commits.

Subject Faculty Name: Prof. Arijit Dutta